User Manual — Halalan Thayiban System

Not just Halal but Goodness

Live system (prototype, invented data):
https://red1oon.github.io/BIMCompiler/halal/

Full paper: https://red1oon.github.io/BIMCompiler/HalalTrace/

Other languages: Bahasa Melayu —
https://red1oon.github.io/BIMCompiler/halal/manuals/Panduan_Pengguna_Halalan_Thayiban.pdf · All manuals:
https://red1oon.github.io/BIMCompiler/halal/manuals/

Source: the paper "Halalan Thayiban System" (sections 6, 7, 14–17). The prototype uses invented data.

1. What is this system?

"Halalan Thayiban" means lawful and good. A shopper scans one QR code on a pack of frozen meat. The phone shows:

- whether the pack is halal, which certificate covers it, and who issued it;
- where the pack has been, and which container carried it;
- a **Thayiban score**: how the animal was fed, kept and handled, and how the meat travelled.

Veterinary officers, customs officers and halal auditors use the same records to trace a lot backward to the farms and forward to the shops.

Principle: each party shares only what the next party needs to know. The rest stays private and is proven when asked.

2. For the shopper (your phone)

No app to install. The phone camera is enough.

1. **Scan.** Point the camera at the QR code on the pack. The verdict page opens.
2. **Read the verdict.** The page states the verdict, the certificate, the issuer, the validity dates and the plant. The phone checks the signature itself.

3. **See the Thayiban score.** A score out of 100, with a bar for each component. Tap to see what is checked.
4. **Track it like a parcel.** At the top: the status and a six-step bar. Below: every handling step, **newest first**. Tap a step to see who handled it and the evidence.
5. **Check the container and cold chain.** Container number, seal, previous cargo, cleaning record, the content listing with each line's certificate, and a temperature chart for each leg against the limit.
6. **Report or review.** Photograph the pack, rate it, pick a type and the shop, and write a note. The message carries the pack link and the verdict shown at scan time. Send it through the phone's share sheet (WhatsApp, email or any app), or just save it.

Offline. The report is stored on the phone first, so it works without a signal. The system sees nothing unless you send it. The status stops at "Shared", because a share sheet cannot tell us more.

3. For the dock worker

Each carton is scanned onto the container. Rule R1 checks it on the spot.

1. Scan the carton. If its certificate is valid for a halal container, it is accepted and the count goes up (for example "carton three of eight").
2. If the carton does not belong (for example pork in a halal container), it is **refused on the spot**. The carton is set aside and the event is logged.
3. The worker cannot skip the refusal. The halal auditor sees the alert on the console at once.

4. For the console (plant, auditor, authorities)

The plant's own app and the authority views share one generic grid.

- **Dashboard:** alerts, including refused scans.
- **Containers:** the content listing against certificates.
- **Recall:** one search finds packs, cartons, pallets, shops and the shoppers who scanned a pack.
- **Veterinary service:** records inspections, holds and releases a lot with one click. The verdict changes at once for every pack in that lot.

- **Customs:** compares the declaration with the manifest and the scans, then releases or holds.
- **Authorities** can ask for a lot to be revealed. The owner discloses the secret salt, and the hash must match the manifest. A wrong salt reveals nothing. Every reveal is logged.

5. Who does what

Party	Job
Farm	Tag animals; scan the tag at movement; keep feed, treatment and barn records
Abattoir and packing plant	Print GS1 labels; scan cartons onto pallets and containers; run the dock rules; send the manifest
Halal certifying body	Make certificate number, scope, plants and dates checkable; vouch for plant signing keys
Veterinary service	Record inspections; place and release holds
Customs and border control	Compare the declaration with the manifest; release or hold
Port and carrier	Scan container arrival; share reefer readings
Cold store and distribution centre	Scan pallets in and out; share temperature summaries
Retailer	Scan cartons received; show the promotion
Consumer halal movement	Recognise the verified chain; promote participating products

Every party needs: a phone or handheld scanner, a label printer where labels are made, an enrolled signing key, and the local app. No RFID and no integration with the party's own ERP is needed to start.

6. The Thayiban score

Each lot is scored out of 100 from five components. Each is a plain count of checks passed, with equal weights.

Component	What is checked
Nutrition	feed ration recorded; no unauthorised growth promoters; clean water always available; feed supplier traceable; finishing diet logged
Animal health	vaccinations current; withdrawal periods honoured; veterinary visit in the last 90 days; mortality within the herd norm; pre-slaughter inspection passed
Barn conditions	space per head meets the minimum; ventilation and bedding adequate; shade and shelter; cleaning log; welfare audit within 12 months
Cold chain	share of legs whose readings stayed within the limit
Container hygiene	cleaned after conventional cargo

The shopper sees only the lot score. A farm's feed, treatment and barn records are private. A farm auditor attests the component scores, and the lot score is the animal-weighted mean of its farms. An authority can ask for by-farm scores, and that reveal is logged.

Note. The criteria, weights and thresholds are a **proposal**. What counts as good husbandry is for veterinarians, religious scholars and standards bodies to set, not for this prototype. The sample scores are invented.

7. Languages

The prototype is available in English, Thai, Malay, Arabic (right to left) and Mandarin. Choose a language from the language menu on the page. The manual icon beside it opens this manual and its other-language versions.

8. Limits

- All data in the prototype is invented.
- The Thayiban score is a proposal, not yet endorsed by any authority.
- Cost and effort of adoption have not been measured.
- Open questions (certificate registry access, key trust, trace cache, hosting) wait on the parties concerned.